

Hasan Bukhari

Computational Biology • Bioinformatics • Research Engineering

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EDUCATION

University of Southern Mississippi

B.S. in Computer Science; Minor in Biological Sciences

GPA: 4.00 • President's List • Academic Excellence Scholar

Beaconhouse College Program

Cambridge A Levels: 4 As in Biology, Chemistry, Mathematics, and Physics

Expected Dec 2028

Hattiesburg, MS

May 2024

Karachi, Pakistan

RESEARCH & TECHNICAL EXPERIENCE

Independent Computational Biology Researcher

2026 – Present

CancerShift and Chromatin Contact Prediction Across Chromosomes

Independent

- **CancerShift:** Developed a leakage-controlled multicancer classification pipeline spanning TCGA, GEO, and CPTAC-GDC cohorts; achieved 99.59% accuracy on a blind microarray cohort and 98.24% on external RNA-seq data.
- **Chromatin contact prediction:** Designed and trained A17, a residual MLP predicting Hi-C contact structure from seven epigenomic tracks under strict chromosome-level holdouts and a locked test set.
- Found that relative chromatin structure generalized across unseen chromosomes more consistently than absolute contact-map scale; documented failed target-blind calibration and negative results in a manuscript in preparation.

Undergraduate Researcher, Ichthyology Lab

Spring 2026 – Present

University of Southern Mississippi

Hattiesburg, MS

- Analyze chromosome-scale *Fundulus* genome assemblies across two related species and 24 pseudochromosome labels to evaluate synteny, correspondence, and candidate fusion/fission patterns.
- Build reproducible Python workflows, alignment-based correspondence matrices, and technical reports translating genomic methods, limitations, and next-step recommendations for lab review.

Data Analyst Intern, Health, Safety & Environment

Jul 2026 – Aug 2026

Engro Technical Services

Karachi, Pakistan

- Built executive Power BI analytics over 130,000+ incident, near-miss, inspection, audit, waiver, and corrective-action records.
- Developed Python, Power Query, and FastAPI pipelines for risk scoring and scheduled reporting; automated daily change monitoring across approximately 1,750 engineering and HSE files.

Student Worker, Operations Technology & Event Support

Jun 2024 – Present

Thad Cochran Center / Union Operations, University of Southern Mississippi

Hattiesburg, MS

- Coordinate technical support and event operations; built SetItUp and UnionInventoryPro for room requests, layouts, equipment tracking, and recurring inventory audits.

Software Engineering Intern

Aug 2023 – Feb 2024

Shahid I.T. Consultants

Karachi, Pakistan

- Automated recurring record-update workflows in Python, reducing manual workload by 30%; supported integration testing and deployment documentation.

SELECTED RESEARCH SOFTWARE

BioLitGraph — Biomedical Literature Graph Explorer | Python, Flask, PubMed/PubTator, NetworkX | [GitHub](#)

2026

- Built a biomedical literature-mining application that retrieves papers, normalizes biomedical entities into relationship graphs, preserves paper-level evidence, and exports analysis tables as CSV/JSON.

PhyloPlayground — Interactive Phylogenetics Toolkit | Python, Streamlit, NumPy, Plotly | [GitHub](#)

2025

- Implemented Neighbor-Joining and UPGMA phylogenetic inference, JC69/K2P evolutionary distance models, FASTA parsing, and bootstrap branch support with interactive tree visualization.

QTL Analysis Toolkit | Python, pandas, Matplotlib | [GitHub](#)

2025

- Implemented LOD scans, permutation-based significance thresholds, significant-marker exports, and publication-ready statistical-genomics plots.

TECHNICAL SKILLS

Programming & Modeling: Python, R, SQL, C++; PyTorch, scikit-learn, pandas, NumPy, SciPy, Biopython

Computational Genomics: RNA-seq, microarrays, Hi-C, ATAC-seq, epigenomics, statistical genomics, synteny, QTL analysis, phylogenetics

Research Engineering: Linux/WSL, Git/GitHub, FastAPI, Flask, Power BI, DAX, Power Query, Excel

Data Resources: TCGA, GEO, CPTAC, ENCODE, Ensembl, PubMed/PubTator

LEADERSHIP & HONORS

President & Founder, Student Poets Association and Creative Writing Club, USM — Lead arts programming, public showcases, open mics, and club operations.

Selected Honors: Outstanding Cambridge Learner Award; ITC 371 Best Project & Best Assistant Project Manager; UGS Best Art as Research Project (2026); Best Arts Showcase for Poetry (2025, 2026); President's List.